

Clustering applied to EGMS InSAR data: the case of the Alqueva dam

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Abstract

Understanding and predicting the behaviour of geotechnical structures is still a complex challenge. Traditional methods often fail to fully capture the complexities of these structures, which include challenges such as slope instabilities, dam behaviour, and landslides. To address these issues, significant advances have been made in Earth observation techniques, such as radar interferometry and satellite imagery, as well as in machine learning tools. These developments have provided new opportunities for monitoring and predictive modelling. This review explores the application of Earth observation techniques, such as InSAR, for detecting ground displacement and deformation. It also examines the role of machine learning in analysing large datasets, detecting patterns and anomalies, and predicting potential geotechnical failures. Despite these advancements, challenges remain, particularly regarding the reliance on high-quality data and the integration of advanced algorithms into practical applications.

1 Introduction

1.1 Contexto

Geotechnical structures, particularly large-scale dams, are essential for global energy and environmental management. Ensuring these structures are safe is vital to prevent disasters. Traditionally, monitoring has focused on "observing the main actions and characterising the quasi-static structural response" (Mata et al., 2023). These standard methods rely on physical inspections and instruments installed directly on the dam (in-situ instrumentation).

Since 1967, the most common way to analyze this data has been the Hydrostatic-Seasonal-Time (HST) model, which looks at how water levels and air temperature affect the dam (Salazar et al., 2017). However, as dams become larger and more complex, these traditional methods are becoming "increasingly insufficient" (Hariri-Ardebili et al., 2023).

A major challenge is that many instruments are "placed in hostile environments, where humidity, temperature change, and freeze" test the limits of mechanical and electronic tools (De Sortis & Paoliani, 2007). Because these sensors can fail or be hard to reach, there is a strong interest in satellite-based Earth Observation (EO). Specifically, a technique called Multi-Temporal InSAR (MT-InSAR) has revolutionized how we monitor ground and structure movements.

Unlike traditional sensors that must be physically attached to the dam, MT-InSAR uses radar signals to measure displacements from space with millimetre-level precision (Cernuto et al., 2026; Xie et al.,

2022). Because it works in any weather or lighting condition, this technology provides a way to monitor dams from a distance, even in remote areas. These satellite-based frameworks offer a more robust and continuous way to ensure safety, acting as a powerful complement to traditional monitoring.

(Mata et al., 2023) "Traditionally, monitoring systems of large concrete dams have been based on observing the main actions and characterising the quasi-static structural response.

(Hariri-Ardebili et al., 2023) "Traditional dam safety methods, based on visual inspections and manual monitoring, have long been the standard for ensuring the stability and safety of dams. However, as the scale and complexity of dam projects have increased, these methods have become increasingly insufficient."

(Salazar González et al., 2017) "Linear regression is the simplest statistical technique, appropriate to reproduce certain phenomena. It is also the basis of the most popular data-based behaviour model in dam engineering: the Hydrostatic-Season-Time (HST). It was first proposed by Willm and Beaujoint in 1967."

(Ruiz-Armenteros et al., 2023) "Periodic in-person inspections by specialized technicians are conducted to maintain the facilities, ranging from daily to annual visits.

(De Sortis & Paoliani, 2007) "Actually it is not easy to get reliable measures by means of instruments often placed in hostile environments, where humidity, temperature change, freeze and uncomfortable access put to the test any kind of mechanical and electronic tool." isto pode ser útil para justificar a framework e ser utilizada em áreas mais remotas

(Ziemer et al., 2025) "The deformation behavior of dams is essentially dependent on exogenous factors, such as the air temperature and water level (i.e., water pressure) of the reservoir, as well as longterm trend components (i.e., time or the static of the structure) [1]. The monitoring program of dams often comprises a variety of procedures to guarantee their safe operation. These procedures, among others, are based on the expected deformation behavior of the dam and include pendulum systems and trigonometric measurements [2]. To test the reliability of the measurements, predictions are made regarding the expected deformation. For pendulum data, this is usually performed on a daily or weekly basis, while trigonometric field campaigns are only conducted a few times a year. Current methods utilize multiple linear regression (MLR) models to predict the expected deformation of a dam based on the aforementioned variables [1,35]. However, these methods may encounter challenges in adequately predicting the deformation since they rely only on a linear combination of temperature and water level daily means. Furthermore, deformation data collected through pendulum systems or trigonometry represent only a specific point on the dam or are measured infrequently due to time-consuming field campaigns, which limits their availability in space or time [69]."

1.2 Avanços Insar e EO

Recent advancements in Multi-Temporal InSAR (MT-InSAR) have revolutionized geotechnical monitoring, offering millimetre-level precision regardless of lighting or weather conditions (Cernuto et al., 2026; Xie et al., 2022). Within the InSAR family, specialized approaches such as Persistent Scatterer (PSInSAR) and Small Baseline Subset (SBAS) have gained prominence for their ability to detect subtle ground movements (Hariri-Ardebili et al., 2020). However, structural monitoring of dams using MT-InSAR is not yet a widespread application (Ruiz-Armenteros et al., 2018). Nevertheless, previous achievements suggest these techniques can support warning systems by adding redundancy at a low cost.

(Marchamalo-Sacristán et al., 2023) ""

1.3 Oportunidade

A major technological opportunity emerged in 2022 with the launch of the European Ground Motion Service (EGMS). This service is part of Copernicus, the European Union's Earth Observation program, and is supported by the European Space Agency (ESA), the organization responsible for developing and launching the satellites.

The EGMS provides standardized, open-access displacement data derived from the Sentinel-1 satellite constellation. This data complements on-site monitoring by offering "broader spatial coverage and continuous monitoring capabilities" (Ruiz-Armenteros et al., 2023). By examining the temporal patterns in these satellite time series, it is possible to identify long-term trends or seasonal variations that are critical for assessing risks. At the same time, Artificial Intelligence (AI) models are becoming more sophisticated, opening "fresh opportunities for dams digital and intelligent development" (Cao et al., 2025).

(Ruiz-Armenteros et al., 2023) "In addition to on-site monitoring techniques, the EGMS complements these methods by offering broader spatial coverage and continuous monitoring capabilities. It assists in identifying areas of interest for further investigation and validation, allowing dam operators and authorities to allocate their resources effectively."

(Ruiz-Armenteros et al., 2023) "The EGMS generates time series data that illustrate the evolution of ground motion over a specific period. By examining the temporal patterns of ground displacement, it becomes possible to identify longterm trends or seasonal variations in the dam's behavior and its surroundings. This information is valuable for understanding the dam's stability and assessing any changes that may pose a risk."

(Cao et al., 2025) "After decades of research, AI in particular are becoming increasingly sophisticated. Theses open up fresh opportunities for dams digital and intelligent development (Sun et al., 2018; Zhang et al., 2019)."

(Ziemer et al., 2025) "we demonstrated considerable improvements in predicting dam deformations and evaluating their drivers"

(Ziemer et al., 2025) "By utilizing satellite-based methods such as Persistent Scatterer Interferometry (PSI), dam deformations can be monitored with millimeter precision in extensive spatial and temporal coverage. Freely available multi-temporal synthetic aperture radar interferometry (MT-InSAR) data provided by ESAs Copernicus Sentinel-1 (S-1) mission constitute a considerable advancement. The implementation of S-1 PS time series into nationwide ground motion services, such as the German Ground Motion Service (Bodenbewegungsdienst Deutschland, BBD), opens up new possibilities for deformation monitoring on a large scale."

1.4 Estudos prévios em barragens

(Ziemer et al., 2025) "To date, only a few studies have concentrated on predicting dam deformations [9,22], underscoring the need for further research in this field."

Several studies have explored InSAR displacement series for dam monitoring. Projects like SIAGUA in Spain have focused on using Copernicus data as routine tools for dam safety assessment (Ruiz-Armenteros

et al., 2023). SARClust tool have introduced semi-automatic clustering of InSAR time series to group points with "homogeneous behavior" (Roque et al., 2023). However, many of these efforts focus on retrospective analysis rather than continuous operational safety frameworks.

(Roque et al., 2023) "In this study, a new tool for semiautomatic analysis of InSAR displacement time series is presented, called SARClustClustering of InSAR displacement time series. The tool forms clusters of InSAR measurement points with homogeneous behavior. This means the points in each cluster present similar displacement time series among themselves, but different from the displacement time series of the points in other clusters."

(Ruiz-Armenteros et al., 2023) "The SIAGUA project, funded by the Spanish Ministry of Science and Innovation, focuses on applying the Copernicus land monitoring program and other satellite radar sensors as routine tools for monitoring dams, large ponds, and their surroundings. EGMS is used to provide initial deformation data to dam managers, enabling surveillance of the infrastructures under their responsibility."

(Ruiz-Armenteros et al., 2023) "The SIAGUA project in Spain utilizes EGMS to monitor dams and surrounding areas, providing surveillance information to dam managers."

(Ruiz-Armenteros et al., 2018) "Structural monitoring of dams using MT-InSAR is not yet a widespread application and only a relative few number of studies have been conducted during the past 15 years for structural health monitoring of dams using SAR data12-24. However, these achievements allowed to conclude that MT-InSAR techniques have the potential to support the development of new and more effective means of monitoring and analyzing the health of dams and add redundancy, at low cost, to support and assist warning systems."

1.5 Limitações (referidas por outros autores)

Despite these benefits, practical integration remains challenging. Machine Learning (ML) models, though often more capable than HST of reproducing non-linear effects, typically focus on specific case studies or arch dam typologies, which represent only 5% of dams worldwide (Salazar et al., 2021). Furthermore, the technical nature of radar data presents a barrier; the "practical utility of solely LOS (Line-of-Sight) deformation is limited for users unfamiliar with InSAR," often leading to confusion among dam managers accustomed to horizontal and vertical displacement (Ruiz-Armenteros et al., 2023). Additionally, users must be cautious of EGMS data accuracy, as discrepancies compared to ground truth can sometimes reach 510 mm/yr due to atmospheric or geometric artifacts (Crosetto et al., 2025).

(Salazar et al., 2021) "There is a growing interest in the application of innovative tools in dam monitoring data analysis. Although only HST is fully implemented in engineering practice, the number of publications on the application of other methods has increased considerably in recent years, specially NN. It seems clear that the models based on ML algorithms can offer more accurate estimates of the dam behaviour than the HST method in many cases. In general, they are more suitable to reproduce non-linear effects and complex interactions between input variables and dam response. However, most of the published works refer to specific case studies, certain dam typologies or determined outputs. Many focus on radial displacements in arch dams, although this typology represents roughly 5 of dams in operation worldwide."

(Salazar et al., 2021) "From a practical viewpoint, data-based models should also be user-friendly and easily understood by civil engineering practitioners, typically unfamiliar with computer science, who have the responsibility for decision making."

(Salazar et al., 2017) "The most popular data-based approach for dam monitoring analysis is the hydrostatic-seasonal-time (HST) model. It was first proposed by Willm and Beaujoint in 1967 [9] to predict displacements in concrete dams, and has been widely used since."

While EGMS provides a continental-scale baseline, its application to specific infrastructure like dams must account for inherent InSAR limitations, such as...:

(Crosetto et al., 2025), "Some of the results show a good agreement between EGMS and the ground truth, both in terms of deformation velocity (absolute differences below 12 mm/yr) and time series (standard deviation of differences below 5 mm). However, this cannot be said generally. In some cases, the differences in the deformation velocities (EGMS vs. ground truth) are far from the millimetre-per-year precision, which is sought. For instance, differences in the range of 510 mm/yr can be found. The same is true for the time series, where biases up to 1020 mm or peaks of the same order of magnitude occur."

(Ruiz-Armenteros et al., 2023) "However, utilizing InSAR technology requires specialized knowledge of various processing algorithms and significant computational resources."

1.6 Research Gap

A critical research gap persists where satellite remote sensing, machine learning, and traditional monitoring operate as "isolated islands" with minimal synergy. The main issue is the lack of integration between these three areas:

- (i) The EGMS island provides regional displacement data, yet it remains abstract for dam managers due to the complexity of Line-of-Sight (LOS) geometry and atmospheric noise (Ruiz-Armenteros et al., 2023);
- (ii) The Machine Learning island offers powerful modeling capabilities, but often lacks the "engineering judgement" and physical grounding required for high-stakes safety decisions (Salazar et al., 2021);
- (iii) The Traditional Monitoring island provides the "ground truth" through high-precision *in-situ* instruments like pendulums, but is limited by its inability to capture the wide-area behavior of the structure and its surrounding environment.

This separation makes it difficult to turn complex satellite data into clear safety information that engineers can use. A major technical problem is that the standard EGMS "Ortho" data uses a 100-meter grid. This resolution is too coarse to show the small, local movements of a specific dam structure (Ruiz-Armenteros et al., 2023). To overcome this, integrating data from both ascending and descending satellite orbits is essential. By combining these two perspectives, we can move beyond the 100-meter limitation and more accurately detect structural movements. This approach allows us to transform raw radar signals into the vertical and horizontal displacements that dam managers actually need. There is, therefore, an urgent need for a new framework that brings these three areas together, finally closing the communication gap between satellite experts and civil engineers.

(Ruiz-Armenteros et al., 2023) "However, the Ortho products, which represent the vertical and east-west deformation, are not useful for this dam due to its relatively small size. A 100meter resample is too large to accurately characterize the vertical and east-west deformation patterns, as it does not capture local movements. Moreover, the Ortho products from the EGMS are not suitable for comparison with on-site measurements such as leveling over the dam's crest or berms." isto é muito importante para justificar a importância da combinação das órbitas ascendente e descendente

(Ruiz-Armenteros et al., 2023) "While the EGMS is highly beneficial for dam managers, the practical utility of solely LOS deformation is limited for users unfamiliar with InSAR, as the deformation is projected in the LOS direction. Dam managers are accustomed to working with deformations in the horizontal and vertical directions, and the deformation in the LOS direction can lead to confusion.

1.7 Solução

To address this challenge, this paper proposes the SAFE-DAM (Satellite Automated Framework for Engineering of Dams). This framework integrates EGMS data with Machine Learning clustering algorithms to transform massive satellite datasets into operational safety indicators. Unlike purely theoretical studies, this work validates the SAFE-DAM methodology using a dataset from the Alqueva Dam. By cross-referencing InSAR-derived clusters with *in-situ* measurements, this research demonstrates how satellite monitoring can complement traditional methods, ensuring that "engineering judgement based on experience" remains central to interpreting automated results (Salazar et al., 2021).

2 Methodology (new)

This study focuses on the Alqueva Dam, a large concrete arch-gravity dam structure. An aerial perspective of the dam and its surrounding reservoir is presented in Figure 1, providing context for the structural geometry and the areas analyzed in this work.



Figure 1: Views of the Alqueva Dam from different perspectives.

2.1 Displacements calculation and validation

The methodology utilizes calibrated ascending (ASC) and descending (DESC) InSAR datasets from the European Ground Motion Service (EGMS) Level-2b product, representing ground displacements along the satellite Line-of-Sight (LOS). To deliver a physically meaningful reference frame for structural health monitoring (SHM), these LOS measurements were decomposed into vertical (d_V) and horizontal east-west (d_H) components.

Rather than relying on single point-to-point pairing, which is highly sensitive to local phase noise, a robust *Spatial Binning* approach was implemented. The study area was discretized into a regular grid with a spatial resolution of $50\text{ m} \times 50\text{ m}$, optimized via a multi-parameter grid search. This resolution successfully balances the spatial density of Persistent Scatterers (PS) over the dam body and adjacent

slopes while preserving the localized geodynamic signal required to monitor both concrete structures and geotechnical embankments.

2.2 Decomposition of Line-of-Sight displacements

The data processing workflow, encompassing spatial filtering, grid aggregation, temporal synchronization, and geometric decomposition, was fully developed in Python utilizing high-performance geospatial and scientific libraries.

Spatial filtering and Grid Aggregation

The initial spatial distribution of the EGMS calibrated points for both geometries was filtered to the dam and its surroundings, defined by fixed coordinate limits in the ETRS89-LAEA (EPSG:3035) projection ($1755050 \leq \text{Northing} \leq 1955850$; $2692250 \leq \text{Easting} \leq 2803250$).

To mitigate spatial non-overlap between ascending and descending passes, all active PS points falling within each 50 m grid cell were aggregated monthly by calculating their arithmetic mean. Consequently, the representative coordinate of each grid cell represents the geometric center of gravity of its constituent scatterers, rather than a single operational radar target.

Temporal synchronization

Since ascending and descending passes are acquired on distinct orbits and calendar dates, a standardized monthly time-series vector was established. Linear interpolation was applied to synchronize both geometries to the same temporal epochs, ensuring that the subsequent algebraic decomposition represents simultaneous structural states.

Trigonometric Decomposition via Cramer's Rule

Let d_{ASC} and d_{DESC} be the temporally synchronized and spatially aggregated LOS displacements for a given grid cell. Let θ_A, θ_D represent the mean local incidence angles, and α_A, α_D the respective satellite track azimuth angles retrieved from the EGMS metadata.

The projection of the true vertical (d_V) and horizontal east-west (d_H) displacements onto the radar lines-of-sight is governed by the following unit directional vectors:

$$u_{E,A} = \sin \theta_A \sin(\alpha_A - 90^\circ) \quad (1)$$

$$u_{UP,A} = \cos \theta_A \quad (2)$$

$$u_{E,D} = \sin \theta_D \sin(\alpha_D - 90^\circ) \quad (3)$$

$$u_{UP,D} = \cos \theta_D \quad (4)$$

This couples the observed LOS displacements into a deterministic 2×2 linear system:

$$\begin{cases} d_{\text{ASC}} = u_{E,A} \cdot d_H + u_{UP,A} \cdot d_V \\ d_{\text{DESC}} = u_{E,D} \cdot d_H + u_{UP,D} \cdot d_V \end{cases} \quad (5)$$

By applying Cramer's Rule, the system matrix determinant ($\det = u_{E,A} \cdot u_{UP,D} - u_{E,D} \cdot u_{UP,A}$) is evaluated. For cell geometries where $|\det| > 10^{-6}$, the explicit decomposition yields the isolated kinematic components:

$$d_H = \frac{d_{\text{ASC}} \cdot u_{UP,D} - d_{\text{DESC}} \cdot u_{UP,A}}{\det} \quad (6)$$

$$d_V = \frac{d_{\text{DESC}} \cdot u_{E,A} - d_{\text{ASC}} \cdot u_{E,D}}{\det} \quad (7)$$

To eliminate reference-frame biases, the resulting time-series for each individual cell were normalized by subtracting the displacement value of the initial acquisition epoch (t_0), establishing a zero-baseline at the start of the monitoring period.

2.3 Environmental Synergy and Parameter Optimization (Grid Search)

To automatically determine the optimal combination of spatial resolution, temporal alignment, and clustering parameters, a comprehensive Grid Search (Sweep) framework was deployed. Candidate configurations were evaluated not only against geometric consistency but face their physical adherence to environmental forcing factors.

For each tested scenario, the decomposed d_V time-series were clustered using Dynamic Time Warping (DTW) distance matrices and hierarchical agglomerative clustering. The resulting structural trends were cross-examined against a multi-variable environmental dataset extracted from the Copernicus ERA5-Land reanalysis model (spatial resolution ~ 9 km) combined with local structural water level records.

A physical coupling score ($Score_C$) was computed for the clusters based on the mean absolute Pearson correlation coefficient (r) across six key environmental drivers acting on the concrete wall and the reservoir slopes:

$$Score_C = \frac{|r_{\text{Temp}}| + |r_{\text{Niv}}| + |r_{\text{Prec}}| + |r_{\text{P_An}}| + |r_{\text{Hum_S}}| + |r_{\text{Evap}}|}{6} \quad (8)$$

where the evaluated forcing variables comprise: (1) Temperature ($Temp$), (2) Hydrostatic Water Level (Niv), (3) Monthly Precipitation ($Prec$), (4) Hydrological Year Accumulated Precipitation (P_An), (5) Soil Moisture (Hum_S), and (6) Evapotranspiration ($Evap$). This physical optimization step ensures that the selected model architecture prioritizes coherent geodynamic responses over mathematical noise.

3 Methodology

This study focuses on the Alqueva Dam, a large concrete dam structure. An aerial perspective of the dam and its surroundings reservoir is presented in Figure 1, providing context for the structural geometry and the areas analyzed in this work.

3.1 Displacements calculation and validation

The methodology utilizes calibrated ascending (ASC) and descending (DESC) InSAR datasets from the EGMS Level-2b product, representing ground displacements along the radar viewing direction. To enable a direct comparison with the EGMS Level-3 (ORTHO) reference, these Line-of-Sight (LOS) measurements were decomposed into vertical (d_V) and horizontal (d_H) components. This decomposition was validated against the independent orthogonal vertical (U) and eastwest (E) displacements provided by the ORTHO product, ensuring the reliability of the derived geodynamic signals across the dam and its surroundings.

For this initial validation phase, both ascending and descending datasets were aggregated into a $100 \text{ m} \times 100 \text{ m}$ grid. This specific resolution was selected to align with the native spatial scale of the ORTHO



Figure 2: Views of the Alqueva Dam from different perspectives.

products, ensuring a consistent comparison between datasets. This aggregation provides a statistical baseline to verify the accuracy of the InSAR decomposition before proceeding to higher-resolution analyses.

Justificação das vantagens em combinar as duas órbitas:

(Cernuto et al., 2026) "Processing ascending and descending geometries separately proved essential to highlight the influence of viewing direction and the complementarity between datasets: phenomena poorly visible in one geometry were often detectable in the other, confirming that integrating both perspectives reduces shadow zones and provides a more complete picture of interferometric observability."

(Ruiz-Armenteros et al., 2023) "While the EGMS is highly beneficial for dam managers, the practical utility of solely LOS deformation is limited for users unfamiliar with InSAR, as the deformation is projected in the LOS direction. Dam managers are accustomed to working with deformations in the horizontal and vertical directions, and the deformation in the LOS direction can lead to confusion."

3.1.1 Decomposition of Line-of-Sight displacements

To derive the vertical (dV) and horizontal (dH) displacement components over the Alqueva dam, the data processing workflow, including temporal synchronization, spatial interpolation, and LOS decomposition, was fully implemented in Python, using a suite of specialized geospatial and scientific libraries.

Spatial filtering

The initial spatial distribution of the EGMS calibrated points for both ascending and descending orbits were filtered to the dam and its immediate surroundings, defined by fixed coordinate limits in the ETRS89-LAEA (EPSG:3035) projection (Figure 3). This map highlights the coverage over the dam's concrete body and adjacent slopes prior to any spatial interpolation or grid aggregation.

Temporal interpolation

Since ASC and DESC acquisitions are collected on distinct dates, a common set of monthly timestamps was defined. Linear interpolation was applied to generate displacement values for both orbits at the same epoch. This ensured temporal consistency and enabled direct comparison between ASC and DESC measurements.

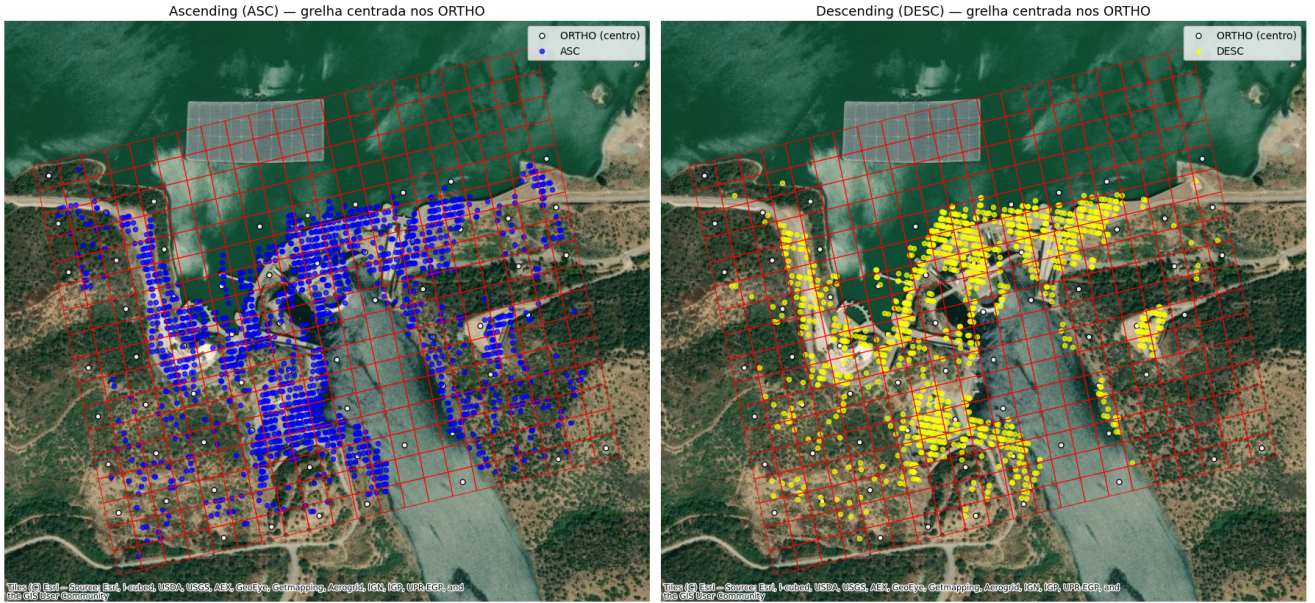


Figure 3: View of the dam and initial ASC/DESC point distribution maps illustrate the area of interest and the spatial coverage of the datasets.

Spatial interpolation

Since ascending and descending PS points do not perfectly overlap in space, an Inverse Distance Weighting (IDW) interpolation was implemented. A search radius of 25 m and a $k = 5$ nearest neighbor limit were applied. The weighting power was set to $p = 2$, following the standard inverse-square law (colocar referência). This spatial resampling is a prerequisite for the LOS decomposition into vertical (d_V) and horizontal (d_H) components, ensuring that the combined measurements represent the same physical location.

The ascending geometry was chosen as the spatial reference for the line-of-sight (LOS) decomposition, with the descending measurements being resampled to the ascending points' locations using the IDW algorithm. This ensures a consistent point-based time-series before spatial aggregation into grid cells.

Computation of geometric parameters (β and γ)

Two auxiliary geometric parameters are defined for each interpolated ascending (ASC) point:

- β (orbital-geometry term) computed as:

$$\beta = \arcsin(\cos(I) \cdot \cos(\text{latitude})) \quad (9)$$

where $I = 98.6^\circ$ is the Sentinel-1 orbit inclination. In this work, β is used as a geometric correction term derived from the satellite orbital inclination and the point latitude. It is not a direct measurement of terrain slope aspect (i.e., not DEM-derived). It accounts for the relative orientation between the satellite orbit plane and the local observation latitude, consistent with the trigonometric formulation implemented in the processing code.

- γ (cross-track tilt) represents the local topographic tilt perpendicular to the radar line-of-sight. Given the relatively flat topography of the area, γ was set to zero ($\gamma = 0$), which simplifies the decomposition while introducing negligible error for this site. ter atenção a isto, pois pode ser uma simplificação pouco rigorosa.

Note: If a DEM-based local slope/aspect correction were applied, β would be replaced or augmented by

the DEM-derived slope aspect, and γ by the DEM-derived cross-track tilt. In this study, those DEM corrections were not used.

Assumptions and implications:

- β interpretation: β is defined as an orbital-geometry term computed from orbit inclination and latitude (not DEM slope aspect). This matches the code implementation and avoids confusion with terrain aspect.
- $\gamma = 0$: Assuming zero cross-track tilt is reasonable for the relatively flat dam area. For steep abutments, γ should be estimated from a DEM. **ter atenção a isto, eventualmente alterar este parâmetro**

Decomposition of Line-of-Sight (LOS) displacements into vertical and horizontal components

(Ruiz-Armenteros et al., 2023) "In this project, we estimate the vertical and east-west displacements by using Detektia's decomposition software, which is based on the nearest neighbors decomposition. Taking the positions of one of the geometries as a reference, each point is paired with the nearest point in the other geometry. The only compliance that they have to fulfill is that the paired points are not separated more than a certain predefined distance threshold."

(Marchamalo-Sacristán et al., 2023) "Once the offsets were estimated and the ascending/descending pairs identified (in the case of ascending and descending dataset pairs), the decomposed movement could be computed using Equations (6) and (7), plotted, and exported."

Let d_{ASC} and d_{DESC} be the LOS displacements from ascending and descending passes, respectively. Let θ_A , θ_D be the local incidence angles (angle between LOS and vertical) for ASC and DESC, and α_A , α_D their respective orbit azimuth (track) angles.

The LOS measurements relate to the vertical and eastwest horizontal components via:

$$\begin{cases} d_{ASC} = -d_V \cos \theta_A + d_H \sin \theta_A \cos(\alpha_A + \beta + \gamma) \\ d_{DESC} = -d_V \cos \theta_D + d_H \sin \theta_D \cos(\alpha_D + \beta + \gamma) \end{cases} \quad (10)$$

Solving this 2×2 trigonometric system yields the vertical and horizontal components:

$$d_V = \frac{d_{DESC} \sin \theta_A \cos(\beta - \gamma) + d_{ASC} \sin \theta_D \cos(\beta + \gamma)}{\cos \theta_A \sin \theta_D \cos(\beta - \gamma) + \cos \theta_D \sin \theta_A \cos(\beta + \gamma)} \quad (11)$$

$$d_H = \frac{d_{DESC} \cos \theta_A - d_{ASC} \cos \theta_D}{\cos \theta_A \sin \theta_D \cos(\beta - \gamma) + \cos \theta_D \sin \theta_A \cos(\beta + \gamma)} \quad (12)$$

These expressions match the implementation used in the processing code (with $\gamma = 0$ for this study).

This formulation ensures that displacements are correctly projected onto the vertical and horizontal reference frame, considering local terrain geometry. The decomposition was applied to each point within the dam area and aggregated into 100m grid cells.

Spatial aggregation into grid cells

The computed d_V and d_H components were aggregated into 100m grid cells centered on the ORTHO points. This spatial aggregation ensured consistency between InSAR-derived components and indepen-

dent orthogonal measurements and facilitated comparison.

Visualization and validation

To evaluate the local performance of the spatial resampling and decomposition process, a representative sample of six grid cells was selected for detailed analysis (Figure 4). For each selected cell, the time-series of vertical (d_V) and horizontal (d_H) displacements were extracted, allowing for a direct comparison between the ASC/DESC IDW-decomposed components and the independent ORTHO reference products.



Figure 4: Spatial distribution of the random selected grid cells used for local validation.

3.1.2 Validation (InSAR vs ORTHO) Accuracy Assessment Metrics

To evaluate the reliability of InSAR-derived displacements, both d_V and d_H components were compared to independent measurements obtained from ORTHO data. For each displacement direction, pairs of co-located observations were extracted, resulting in two datasets (Figure 5):

- d_V^{InSAR} vs. d_V^{Ortho}
- d_H^{InSAR} vs. d_H^{Ortho}

The consistency between InSAR and Ortho displacements was evaluated using some standard statistical metrics, RMSE, MAE, and Pearsons r (Equations (13), (14) and (15)), measuring respectively the average error, absolute deviation, and linear correlation between datasets (Figure 6 and 7).

- Root Mean Square Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (13)$$

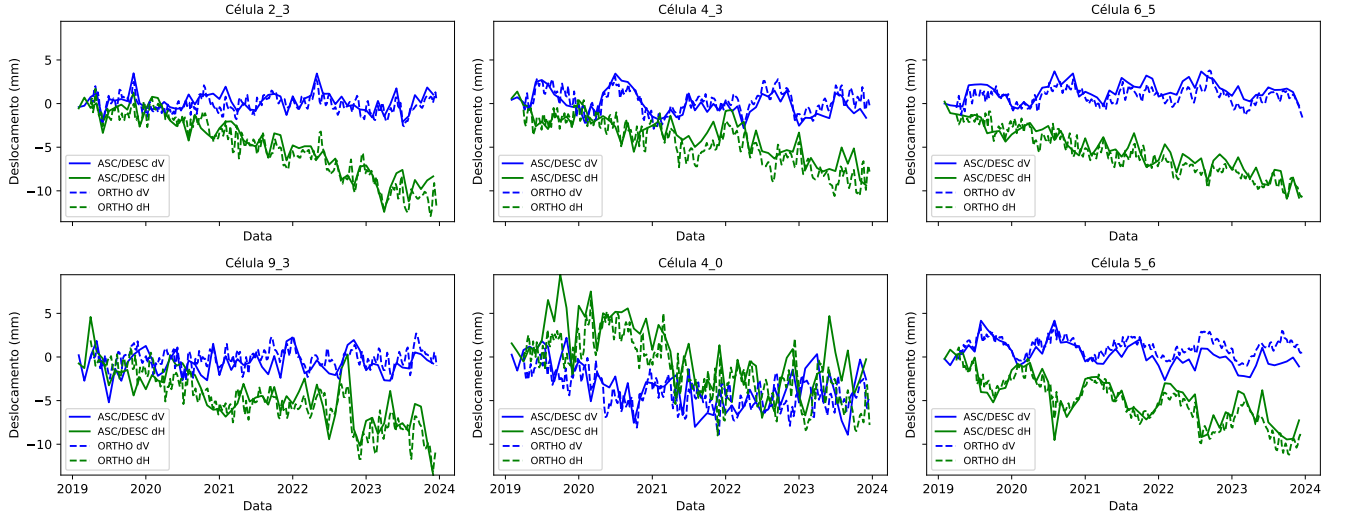


Figure 5: Time-series plots of dV and dH from ASC/DESC IDW compared to ORTHO.

- Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (14)$$

- Pearson Correlation Coefficient (r):

$$r = \frac{\text{cov}(y_i, \hat{y}_i)}{\sigma_y \sigma_{\hat{y}}} \quad (15)$$

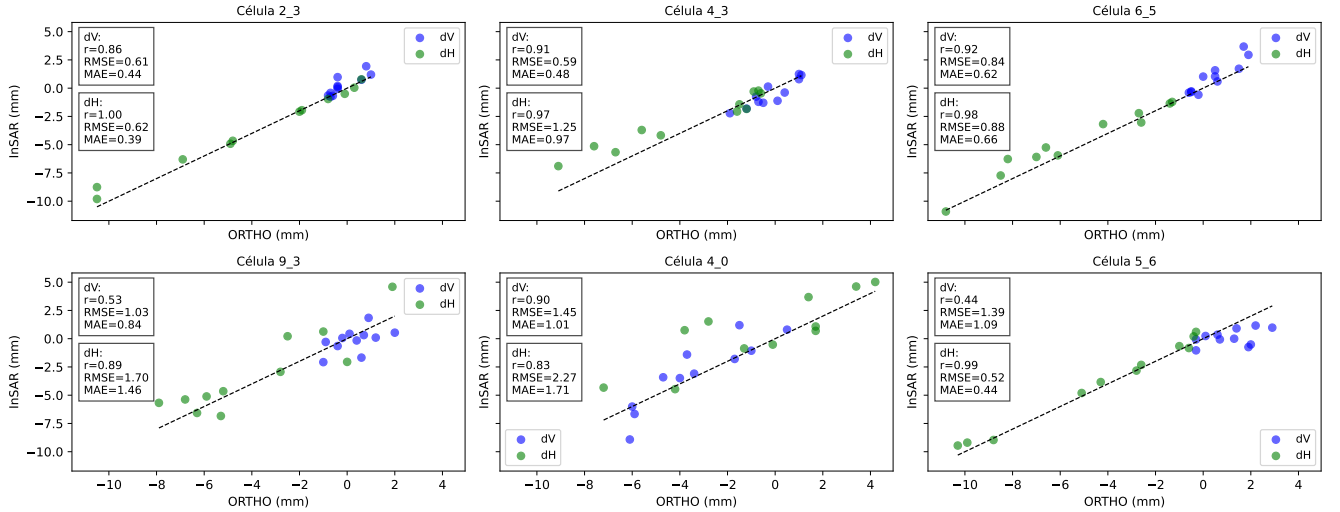


Figure 6: Local-scale correlation analysis for a representative sample of grid cells. Each panel displays the scatter plot between InSAR-derived and ORTHO displacements (dV and dH), including cell-specific error metrics (RMSE, MAE, and Pearson's r).

The global-scale validation (Figure 7) demonstrates a high degree of agreement between the decomposed InSAR components and the ORTHO reference dataset. For the vertical displacement (dV), the analysis yielded a Pearson correlation of $r = 0.86$, with an RMSE of 1.04 mm and an MAE of 0.75 mm. The horizontal component (dH) showed even higher consistency, achieving a correlation of $r = 0.95$, despite a slightly higher RMSE of 1.36 mm and MAE of 0.94 mm. These sub-millimetric average errors ($\text{MAE} < 1$ mm) confirm that the proposed decomposition workflow and the 100 m spatial aggregation effectively minimize local phase noise while preserving the dam's fundamental geodynamic signal. The higher correlation in dH suggests that the decomposition is particularly robust in capturing the East-West seasonal fluctuations.

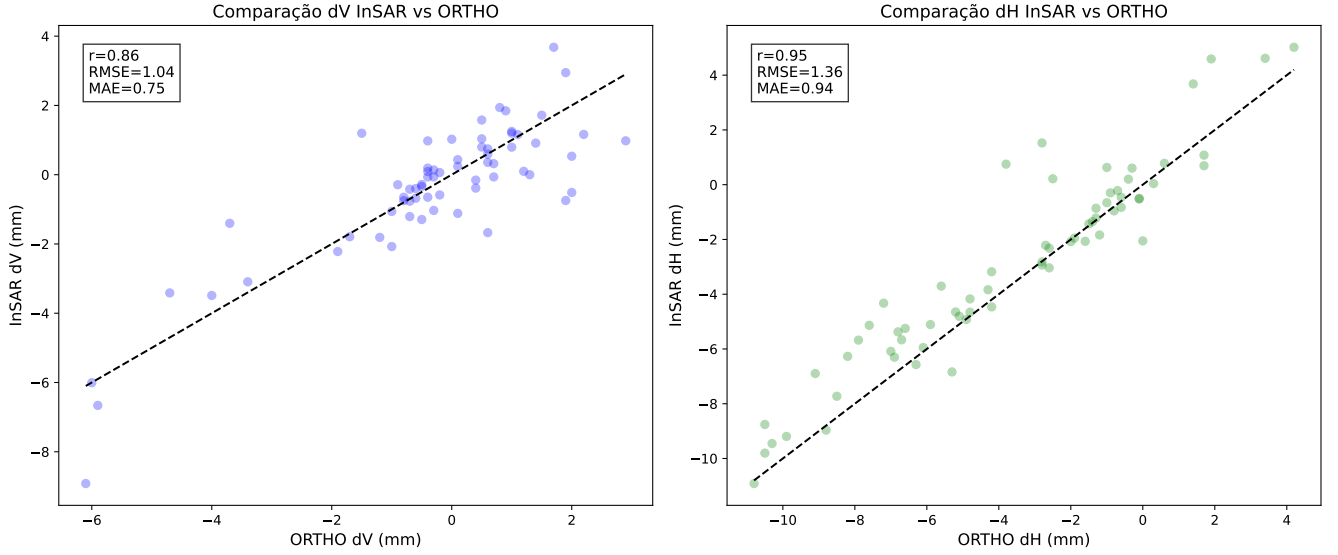


Figure 7: Global-scale validation based on the ensemble average of all cells. This figure correlates the mean structural trend of the dam from InSAR with the mean ORTHO reference.

Although the methodology was rigorously validated for both vertical and horizontal components to ensure the reliability of the decomposition, the subsequent pattern analysis and environmental modeling focus exclusively on vertical displacements (d_V). This choice is motivated by the primary interest in structural settlement and the vertical geodynamic response of the dam’s foundation and abutments.

The high statistical consistency observed at the 100 m scale provides the necessary reliability to apply this decomposition workflow to higher-resolution grids.

3.2 Clustering and Pattern Analysis

3.2.1 Overview

Clustering techniques were applied to identify spatial and temporal patterns of deformation, focusing specifically on the vertical displacement (d_V). These unsupervised machine learning methods enable the grouping of areas with similar deformation behaviour, facilitating the identification of zones with homogeneous displacements, potential structural anomalies, or early indicators of instability.

A grid size of 50 m was implemented as it offers a higher spatial resolution than the standard 100 m EGMS ORTHO grid, allowing for more detailed mapping while maintaining a sufficient number of PS points per cell. This choice follows the rationale that a 100 m resample may be too coarse to accurately characterize local deformation patterns in specific dam structures (Ruiz-Armenteros et al., 2023). Furthermore, as noted by Cernuto et al. (2026), a grid size in the range of 25–50 m serves as a vital compromise: it prevents an overly generalized representation that loses spatial detail while avoiding an excessively fine grid where cells might contain no data due to InSAR point density.

3.2.2 Hierarchical Clustering using DTW Distance

The algorithm adopted was Hierarchical Clustering using Dynamic Time Warping (DTW) distance. The hierarchical algorithm begins by treating each time-series as an individual cluster and successively merges them based on their DTW-based distance, according to a chosen linkage criterion. Unlike conventional Euclidean metrics, DTW accounts for time shifts and non-linear temporal variations, making it particularly suited for deformation signals influenced by seasonal or delayed processes. DTW measures

the similarity between two sequences $X = (x_1, \dots, x_n)$ and $Y = (y_1, \dots, y_m)$ by finding the optimal alignment path w that minimizes the cumulative distance:

$$\text{DTW}(X, Y) = \min_w \sum_{(i,j) \in w} |x_i - y_j| \quad (16)$$

To control the elasticity of the alignment, a warping window constraint was implemented. A window of 0% corresponds to the standard Euclidean distance (point-to-point comparison), while higher percentages allow the algorithm to match shapes that are slightly shifted in time. In this study, the warping window was tested between 0% and 10% of the time-series length. A 1% threshold was selected as the optimal balance to accommodate minor seasonal phase shifts without forcing alignments that would compromise the physical meaning of the clusters.

The linkage method chosen was *Average*. To define the optimal number of groups, the hierarchical structure of the displacement time-series was visualized through a dendrogram (Figure 8). This diagram illustrates the successive merging process based on DTW distances, where the vertical axis represents the degree of dissimilarity between groups. A manual cut-off was applied to the dendrogram to define three distinct clusters. The sensitivity analysis outcomes and the finalized processing parameters for the entire workflow are summarized in Table 1.

Table 1: Summary of Sensitivity Analysis and Final Parameter Selection.

Parameter	Tested Values	Final Choice	Selection Criterion
Grid Size	20, 50, 100 m	50 m	Spatial detail vs. PS density
Temporal Interpolation	-	Linear	Alignment of Ascending and Descending orbits to a shared date
Spatial Interpolation	IDW, Kriging	IDW	Local influence of PS displacement values
IDW Radius (r)	10, 15, 30 m	25 m	Local stationarity of InSAR points
IDW Neighbors (k)	3, 5, 8	5	Preservation of local gradient
IDW Power (p)	1, 2, 3	2	Standard inverse-square distance weighting
Reference Geometry	Asc., Desc.	Ascending	Consistency across multi-platform datasets
Coherence Threshold	0.6, 0.7, 0.8	0.7	Filtering of low-reliability PS points
Linkage Method	Ward, Comp., Avg.	Average	Physical separation of deformation regimes
DTW Window	0% – 10%	1%	Thermal/hydraulic lag compensation

The spatial distribution of the resulting clusters is mapped in Figure 9. The spatial partitioning distinguishes between the structure and the surrounding natural terrain, suggesting that the algorithm successfully captured the differing mechanical responses of the site.

The temporal behavior of each cluster, integrated with environmental variables, is presented in Figure 10. By superimposing the mean cluster signature (solid line) onto the individual cell profiles (grey lines), it is possible to assess the intra-cluster consistency and visually correlate displacement fluctuations with seasonal changes in temperature, reservoir levels, and precipitation.

3.2.3 Temporal Decomposition of Clustered Time-Series using STL

To further explore the temporal dynamics of the displacements, the Seasonal-Trend decomposition using Loess (STL) was applied to the cluster-mean series derived from the DTW-Hierarchical clustering. STL is a statistical method for decomposing time-series into three distinct components: Trend, Seasonal, and residuals. The decomposition follows an additive model (Equation (17)), where the observed time-series (Y_t) is expressed as the sum of these components:

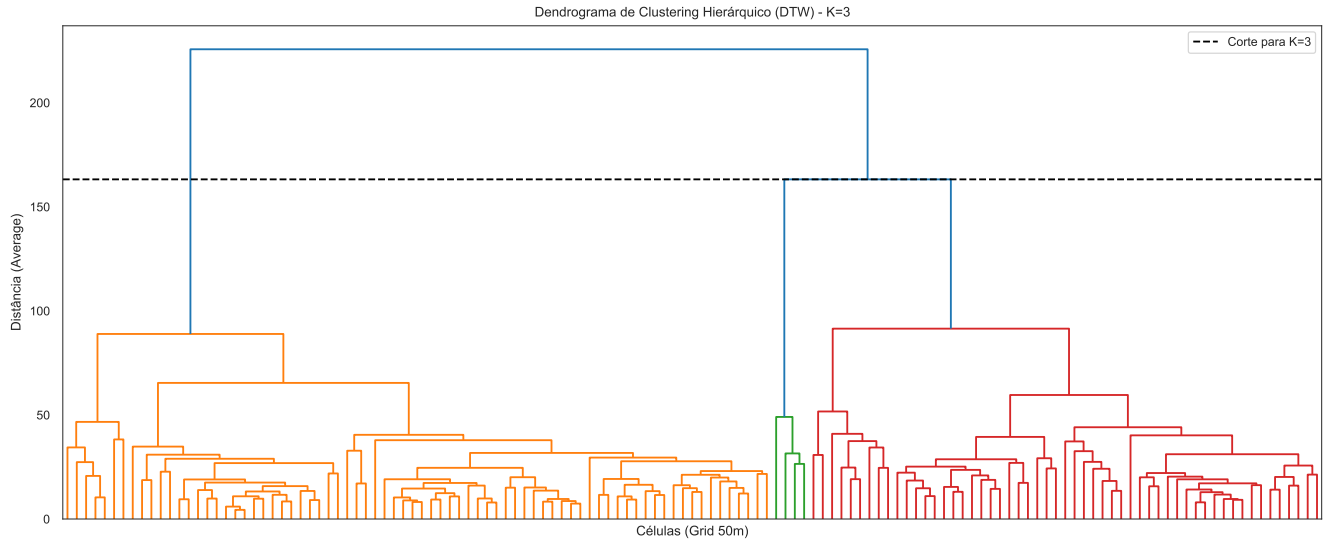


Figure 8: Dendrogram of the hierarchical clustering using DTW distance. The vertical axis represents the merging distance (dissimilarity). The dashed horizontal line indicates the cut-off point to generate 3 clusters.

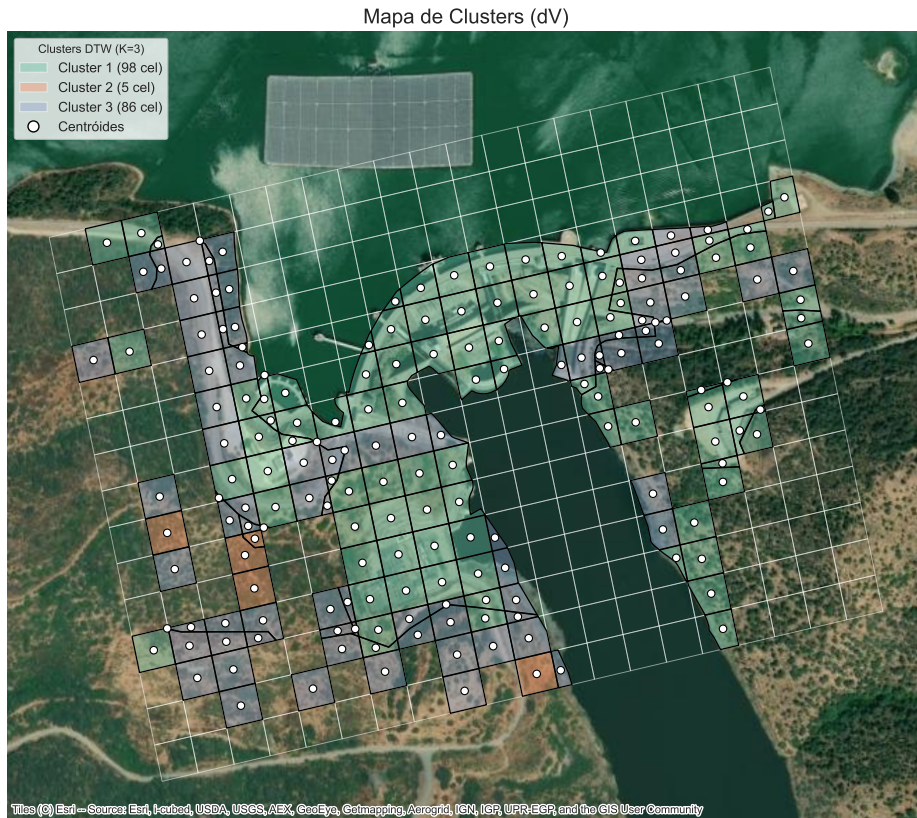


Figure 9: Spatial distribution of displacement clusters at the Alqueva Dam site.

$$Y_t = T_t + S_t + R_t \quad (17)$$

where Y_t represents the observed InSAR displacement at time t , T_t is the trend component, S_t is the seasonal component, and R_t are the residual components.

In the context of dam monitoring, this separation is crucial. The Seasonal component (S_t) captures the cyclical elastic response of the structure and surrounding terrain to annual environmental drivers, such as

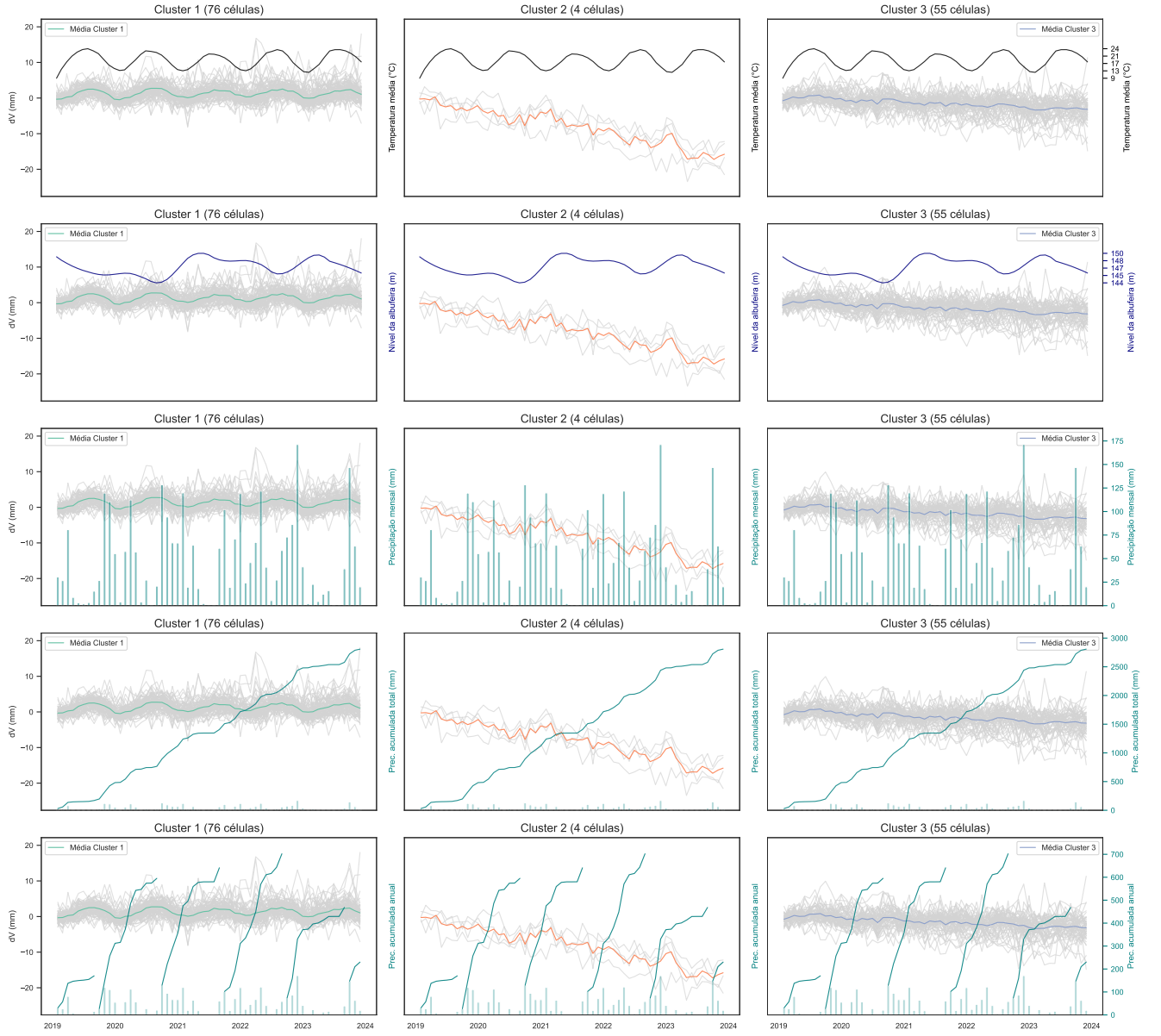


Figure 10: Multivariate analysis of vertical displacement (dV) time series per cluster. For each cluster, external environmental variables: Temperature, Reservoir Level, and Precipitation, are integrated.

temperature fluctuations, periodic reservoir loading, and precipitation patterns. The Trend component (T_t) isolates long-term geodynamic behavior, revealing potential structural settlements or stabilization patterns that are often masked by high-amplitude seasonal oscillations. Finally, residuals (R_t) provide a measure of the signal-to-noise ratio, where low residual values indicate a high degree of fit for the decomposition model.

As shown in Figure 11, the STL decomposition was applied to the vertical displacement (dV) for each cluster. This allowed for a clear distinction between the stable, seasonally-driven behavior of the concrete dam body and the continuous settlement trends observed in the natural terrain and abutment zones.

3.2.4 Multivariate Correlation Analysis

To explore the relationship between deformation patterns and external variables, multivariate pair plots were generated (Figures 12, 13 and 14). This allowed for a simultaneous correlation analysis between the InSAR-derived displacements (dV , dH) and environmental variables (temperature, reservoir level and precipitation). The diagonal shows the distribution of each variable. The lower triangle presents scatter

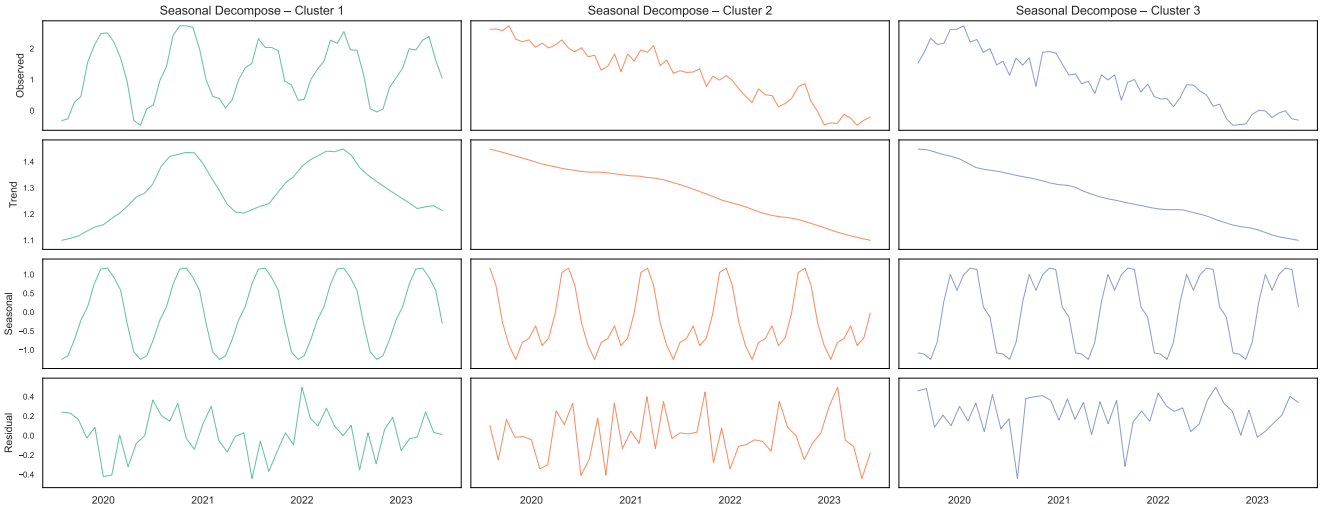


Figure 11: STL decomposition of mean vertical displacement (d_V) per cluster: trend (top), seasonal component (middle), residuals (bottom). Cluster colours correspond to K-Means results.

plots with linear regression lines, while the upper triangle displays Pearson correlation coefficients (r).

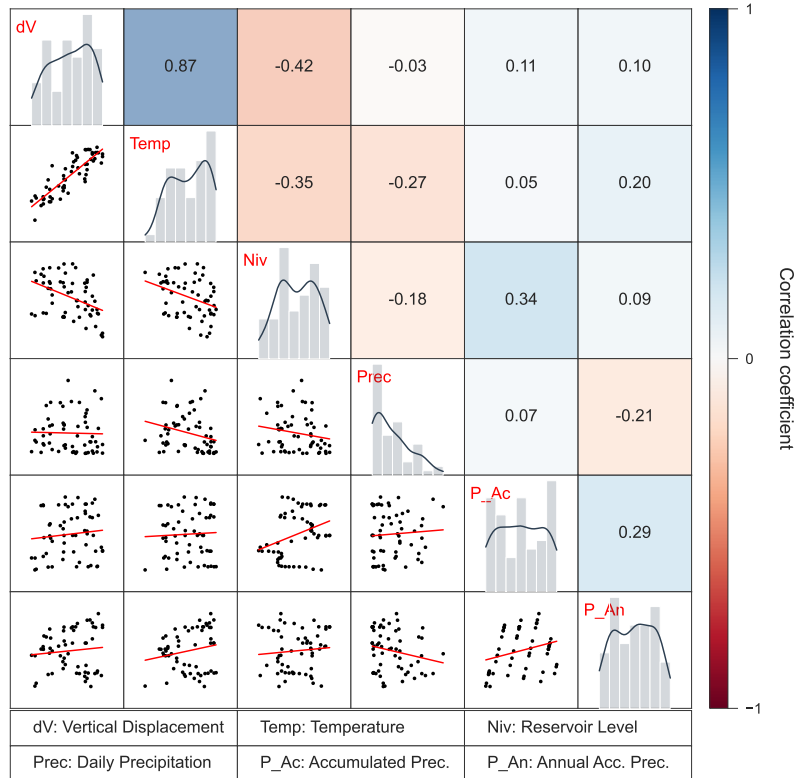


Figure 12: Multivariate correlation matrix for Cluster 1 (Main Dam Body).

3.2.5 Discussion of Results

The spatial and temporal analysis of vertical displacements (d_V) at the Alqueva Dam identified three distinct behavioral patterns. The methodology effectively distinguished between the structural concrete elements and the surrounding natural terrain.

Cluster 1 (Dam body?), characterized by a strong positive correlation with ambient temperature ($r = 0.87$), represents the exposed concrete surface where deformation is primarily driven by thermoelastic

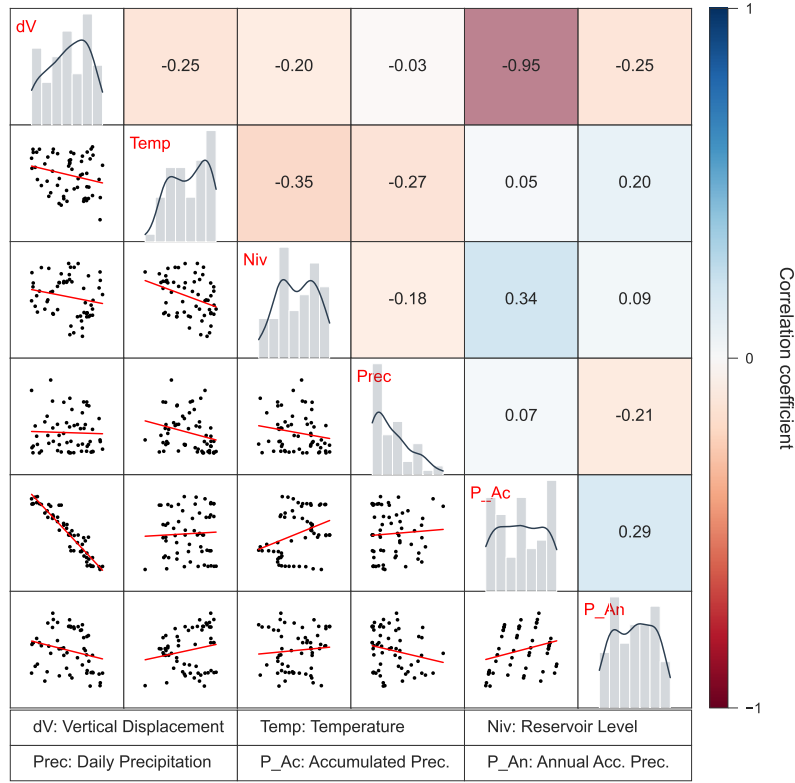


Figure 13: Multivariate correlation matrix for Cluster 2 (Right Abutment/Slopes).

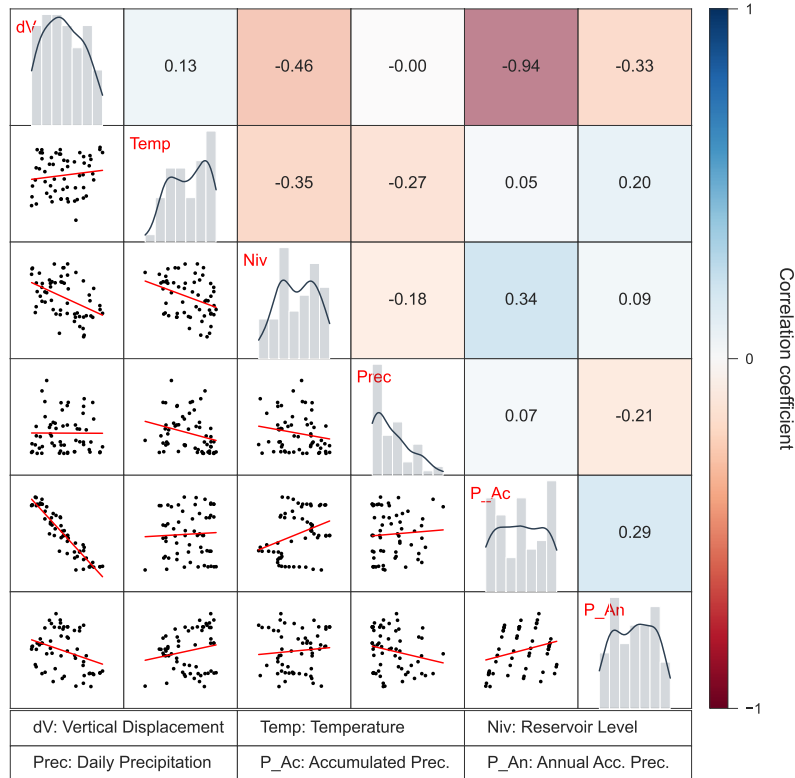


Figure 14: Multivariate correlation matrix (Pair Plot) for Cluster 3 (Left Abutment/Foundation).

expansion and contraction cycles. Furthermore, the moderate negative correlation with the reservoir level ($r = -0.42$) characterizes the mechanical response of the arch-gravity structure to hydrostatic pressure. As the water level rises, the increasing load induces a downstream deflection; due to the dam's

arch geometry and subsequent crest rotation, this motion is captured by the InSAR sensor as a slight vertical settlement (d_V) (embora seja verdade, tentar encontrar referência para isto.).

Cluster 2 (Settlement Trend?) is characterized by a continuous and linear downward trend, visible in both the observed data and the Trend component of the STL decomposition. Interestingly, it shows a near-perfect negative correlation ($r = -0.95$) with Accumulated Precipitation, which may indicate long-term consolidation processes or a delayed hydrological response in specific areas of the abutments or adjacent natural terrain.

Cluster 3 (Geodynamic Response?) represents the terrain surrounding the reservoir. While it maintains a seasonal response to temperature ($r = 0.13$), its Trend component shows a moderate decline over the years. The correlation with the Reservoir Level is negative ($r = -0.46$), suggesting that the filling of the reservoir may induce minor settlements due to hydrostatic loading on the banks.

The STL decomposition effectively isolated the signal from the noise, with residuals generally remaining below 0.4 mm. The seasonal component is consistent across all clusters, but the Trend component clearly differentiates the structural stability of the concrete dam (Cluster 1) from the more persistent movements detected in Clusters 2 and 3.

4 Conclusion

This study demonstrates the effectiveness of integrating InSAR data with machine learning for infrastructure monitoring. The DTW-based algorithm successfully distinguished the cyclical thermoelastic response of the dam (Cluster 1) from localized settlement trends (Cluster 2) and natural terrain responses (Cluster 3).

The strong correlations between displacements and environmental variables validate the physical origin of the InSAR signals. Methodologically, combining STL decomposition with multivariate analysis proved fundamental in isolating millimetric movements, offering a cost-effective and spatially continuous alternative to traditional monitoring displacement systems.

Data Availability

The datasets analysed during this study are publicly available through the European Ground Motion Service (EGMS) platform (<https://egms.land.copernicus.eu/>). To ensure the reproducibility of the results and the validation performed against the ORTHO products, the specific product identifiers used for the Alqueva Dam site are listed below:

- Ascending LOS (Level-2b): EGMS_L2b_052_0848_IW2_VV_2019_2023_1
- Descending LOS (Level-2b): EGMS_L2b_147_0224_IW2_VV_2019_2023_1
- Orthogonal Up (Level-3): EGMS_L3_E27N18_100km_U_2019_2023_1
- Orthogonal East (Level-3): EGMS_L3_E27N18_100km_E_2019_2023_1

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